



Cloud **Terraformation**

FROM CLICKOPS TO
INFRASTRUCTURE AS CODE



Agenda

STORYTIME
THE LANGUAGE
THE PROVIDER
THE ASSIGNMENT
QUESTIONS

Hello from Grand Rapids!

I'm Victor Frye

Your friendly neighborhood developer

Unchallenged #1 Clippy fan

Love internet culture and content

- victorfrye.com/blog
- microsoftgraveyard.com
- shrugman.com

Passion for .NET, React, Azure, Xbox

DevOps specialist at Leading EDJE

Building with Microsoft solutions



About **Leading EDJE**

Technology consultancy and services

- Strategists, innovators, partners
- Craft **bespoke solutions**
- **Positive disruption** of business norm

Real. Fun. Geeks.

FREE lunch and learns

Expert talent

- **Cloud infrastructure**
- Digital and app innovation
- Data and AI
- Project management



The **ClickOps** story

Build application

What infrastructure?

Click, click, deploy

Cross fingers it works

Please do not change it

The background features a light blue gradient with several abstract, soft-edged shapes in shades of blue and purple. These shapes resemble clouds or large, inflated balloons, scattered across the frame. Smaller, solid-colored spheres in similar hues are also present, adding to the abstract composition.

The Answer

CODIFY INFRASTRUCTURE

The background is a light blue gradient with various abstract elements. There are several spheres of different sizes in shades of blue and purple. At the bottom, there are larger, more complex organic shapes in similar colors, resembling liquid or soft blobs. The overall aesthetic is clean and modern.

The Language

INFRASTRUCTURE AS CODE

HashiCorp Terraform

An infrastructure as code tool

Source-available

Cloud-agnostic

Declarative

Stateful

Uses Terraform configuration language (HCL)

Elements include:

- Blocks (e.g., resources, variables, data)
- Arguments (e.g., name = "value")
- Expressions (e.g., name = var.example)


```
resource "azurerm_resource_group" "this" {  
  name      = "rg-example"  
  location  = "Central US"  
}  
  
resource "azurerm_static_web_app" "this" {  
  name                  = "stapp-example"  
  resource_group_name = azurerm_resource_group.example.name  
  location              = azurerm_resource_group.example.location  
}
```

EXAMPLE CONFIGURATION FOR AN AZURE STATIC WEB APP

Write, Plan, Apply



Alternative options



Bicep



az CLI



CloudFormation



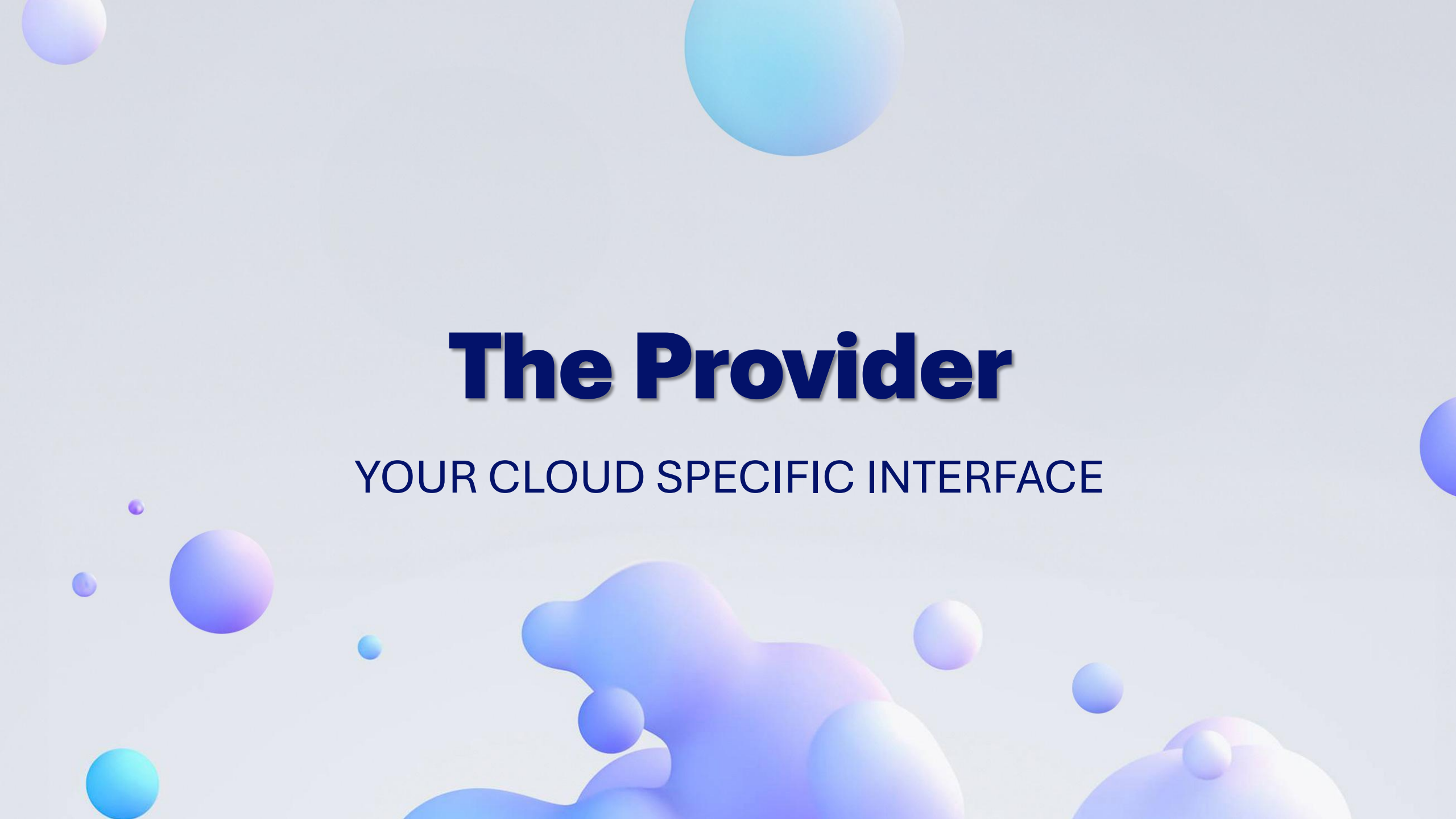
Pulumi



Ansible



PowerShell

The background features a light blue gradient with various abstract elements. There are several spheres of different sizes in shades of blue and purple. At the bottom, there are larger, more complex organic shapes in similar colors, resembling liquid or smoke. The overall aesthetic is clean and modern.

The Provider

YOUR CLOUD SPECIFIC INTERFACE

Multiple providers

AWS

AzureRM

Google

Kubernetes

AzAPI

AzureAD

MSGraph

GitHub

Choose your provider



AzureRM

- Automatic API versioning
- Simplicity
- Comprehensive documentation



AzAPI

- Immediate API access
- Targeted resource updates
- Fine-grained control

Terraforming Azure

ClickOps resources in the Azure portal

Configure workload identity federation with Entra ID

Set up Terraform backend with blob storage

Use template exports for Terraform

Import existing resources into state

Automate CI/CD with GitHub Actions



The background is a light blue gradient with various abstract elements. There are several spheres of different sizes in shades of blue and purple. At the bottom, there are larger, more complex organic shapes in similar colors, resembling liquid or soft clay. The overall aesthetic is clean, modern, and artistic.

The Assignment

READY, SET, GO!

The background features a light blue gradient with several abstract, soft-edged shapes in shades of blue and purple. These shapes resemble clouds or large, inflated balloons, scattered across the frame. Smaller, solid-colored spheres in matching hues are also present, adding to the abstract, modern aesthetic.

Install Terraform

DEMONSTRATION

The background features a light blue gradient with several abstract, organic shapes and spheres in shades of blue and purple. These shapes are scattered across the frame, with some appearing larger and more prominent than others. The overall aesthetic is clean and modern.

Set up identity

DEMONSTRATION

The background features a light blue gradient with several abstract, 3D-rendered shapes. On the left side, there are large, overlapping, rounded blobs in shades of blue and purple. Scattered across the right side are several smaller, smooth spheres in similar colors. The overall aesthetic is clean and modern.

Create backend

DEMONSTRATION

Create backend

The screenshot displays the Microsoft Azure portal interface for a storage account named 'stvftfscus'. The left sidebar contains a navigation menu with options like Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, Partner solutions, Resource visualizer, Data storage (with sub-items Containers, File shares, Queues, Tables), Security + networking, Data management, Settings, and Monitoring. The main content area shows the 'Overview' page with a top bar containing search, Copilot, and action buttons. Below this is a table of essential information including Resource group, Location, Subscription, Subscription ID, Disk state, and Tags. To the right of this table is a table of performance and replication details. At the bottom, there are sections for 'Blob service' and 'File service' settings, each with a table of configurations.

Essentials	
Resource group (move)	rg-1fstate
Location	centralus
Subscription (move)	victorfrye
Subscription ID	95e10bb1-25b8-47dd-b998-267d0af997d2
Disk state	Available
Tags (edit)	Add tags

Performance and Replication	
Performance	Standard
Replication	Locally-redundant storage (LRS)
Account kind	StorageV2 (general purpose v2)
Provisioning state	Succeeded
Created	10/11/2025, 7:47:45 PM

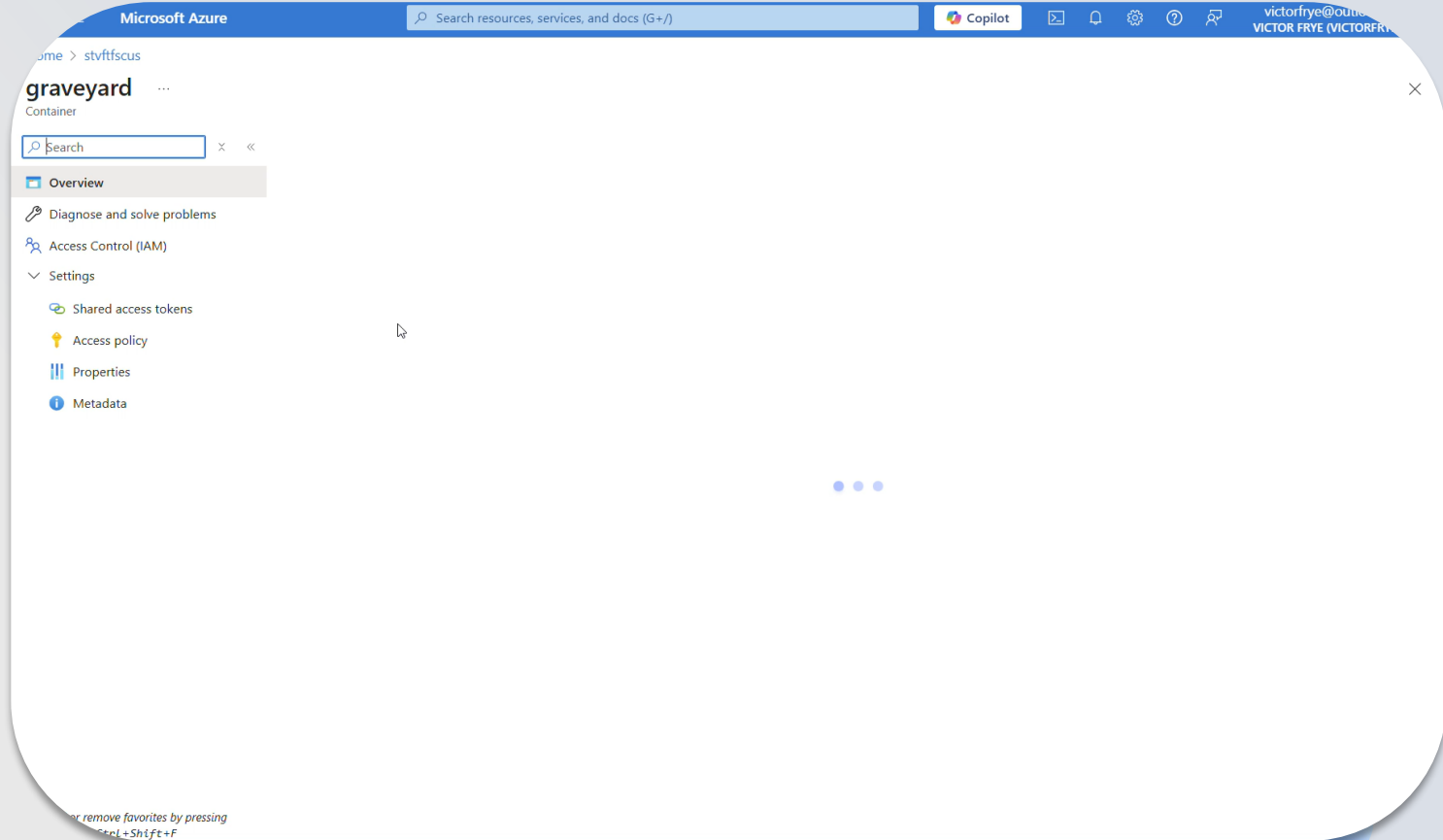
Blob service	
Hierarchical namespace	Disabled
Default access tier	Hot
Blob anonymous access	Disabled
Blob soft delete	Disabled
Container soft delete	Disabled
Versioning	Disabled
Change feed	Disabled
NFS v3	Disabled
Allow cross-tenant replication	Disabled
Storage tasks assignments	None

File service	
Large file share	Enabled

Security	
Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

Networking	
Public network access	Enabled
Private endpoint connections	0
Network routing	Microsoft network routing
Endpoint type	Standard

Configure RBAC



Export Terraform

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

victorfr@outlook.com
VICTOR FRYE (VICTORFR)

Home >

rg-graveyard
Resource group

Export resource groups using Bicep or Terraform | Generate Bicep code to duplicate this resource group. | How to manage these changes more efficiently with deployment tools?

Search

+ Create | Manage view | Delete resource group | Refresh | Export to CSV | Open query | Assign tags | Move | Group by none

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Deployments

Security

Deployment stacks

Policies

Properties

Locks

Cost Management

Monitoring

Automation

Export template

Help

Essentials

Resources | Recommendations

Filter for any field... | Type equals all | Location equals all | Add filter

Name ↑	Type	Location
microsoftgraveyard.com	DNS zone	Global
stapp-graveyard	Static Web App	Central US

Showing 1 - 2 of 2. Display count: auto

Give feedback

The background features a light blue gradient with several abstract, soft-edged shapes in shades of blue and purple. These shapes include large, irregular blobs and smaller, smooth spheres of varying sizes, some of which are partially cut off by the edges of the frame. The overall aesthetic is clean and modern.

Refactor configurations

DEMONSTRATION

The background features a light blue gradient with several 3D-rendered elements. On the left side, there are large, organic, bubble-like shapes in shades of blue and purple. Scattered across the right side are several smaller, smooth spheres in various shades of blue and purple. The overall aesthetic is clean and modern.

Import state

DEMONSTRATION

The background features a light blue gradient with several abstract, organic shapes and spheres in shades of blue and purple. These shapes are scattered across the frame, with a larger, more complex shape on the left and several smaller spheres on the right and bottom.

Build plan

DEMONSTRATION

The background features a light blue gradient with several 3D-rendered elements. On the left side, there are large, overlapping organic shapes in shades of blue and purple, resembling bubbles or soft clouds. Scattered across the right side are several smaller, smooth spheres in various shades of blue and purple. The overall aesthetic is clean, modern, and minimalist.

Apply plan

DEMONSTRATION

The background features a light blue gradient with several abstract, soft-edged shapes in shades of blue and purple. These shapes include large, rounded blobs and smaller spheres, some of which are semi-transparent, creating a modern, minimalist aesthetic.

Automate CI/CD

DEMONSTRATION

The background features a light blue gradient with several abstract, soft-edged shapes in shades of blue and purple. These shapes resemble clouds or large, inflated balloons, scattered across the frame. Smaller, solid-colored spheres in similar hues are also present, adding to the abstract composition.

Did it work?

SHRUGMAN.COM

Thank you

Provide feedback



VICTOR FRYE

616-706-7407

VICTORFRYE@OUTLOOK.COM

VICTOR.FRYE@LEADINGEDJE.COM

LEADINGEDJE.COM

VICTORFRYE.COM/BLOG

LINKEDIN.COM/IN/VICTORFRYE

GITHUB.COM/VICTORFRYE/PRESENTATIONS